

DEMAND INTELLIGENCE

Retail Demand Analysis & Forecasting

POWER BI CASE STUDY

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1. Executive Summary

Understanding demand requires more than tracking total sales over time. Businesses also need to understand how demand changes seasonally, which stores and products contribute most to overall performance, and whether historical patterns can provide a useful view of future demand.

This project analyzes retail demand patterns using historical sales data across **10 stores and 50 products** from **January 2013 to December 2017**. The analysis was developed using **SQL for data exploration and preparation** and **Power BI for data modeling, DAX calculations, analysis, and dashboard development**.

The project was structured around three connected analytical perspectives:

- **Demand Overview** — understanding overall demand trends, growth patterns, seasonality, and monthly performance.
- **Performance Analysis** — identifying the stores and products contributing most to total demand and evaluating how concentrated demand is across the business.
- **Demand Forecasting** — using historical demand patterns to estimate future demand and visualize forecast uncertainty.

The analysis recorded **47.70M in total demand** across the historical period, with overall demand showing a clear upward trajectory alongside recurring seasonal fluctuations. The strongest annual growth occurred in **2014, when demand increased by 15.04% year over year**. Growth later moderated, reaching **3.64% in 2017**, indicating that while demand remained strong, the rate of expansion had slowed.

Monthly analysis revealed a consistent seasonal pattern, with demand generally increasing through the middle of the year and reaching its highest average level in **July at approximately 1.0M**. Demand then moderated toward the end of the year. This recurring pattern provided an important foundation for the forecasting analysis.

Performance analysis showed that demand was relatively broad-based across the business. **Store 02** was the largest individual contributor, generating approximately **6.1M in total demand**, while **Product 15** was the highest-performing product at approximately **1.61M**. The **top three stores contributed 36.5% of total demand**, while the **top ten products contributed 31.4%**, indicating that performance was supported by multiple stores and products rather than being dominated by a single contributor.

The forecasting analysis used historical demand patterns to estimate expected demand for the following 12 months. The forecast continued to reflect the seasonal patterns observed in the historical data, with higher expected demand during the middle of the year. The confidence interval also widened further into the forecast horizon, highlighting the increasing uncertainty associated with longer-term predictions.

Overall, the analysis demonstrates that the business has a **strong and relatively diversified demand base**, supported by recurring seasonal patterns across multiple stores and products. However, the moderation in year-over-year growth and the uncertainty surrounding future demand suggest that historical performance should be monitored alongside forecast results rather than relying on past growth trends alone.

Key Conclusion

Demand has grown substantially over time and remains broadly distributed across the business, but growth has moderated in recent years and future performance should be evaluated alongside seasonal patterns and forecast uncertainty.

The completed Power BI dashboard provides a connected analytical view of historical demand, performance concentration, and expected future trends, supporting more informed demand monitoring and planning.

2. Business Scenario & Objectives

Business Scenario

A retail business operates across multiple stores and products and needs a clearer understanding of historical demand performance, key contributors, and expected future demand.

While total sales data is available, decision-makers need more than headline figures. They need to understand how demand has changed over time, whether growth is accelerating or slowing, which stores and products contribute most to overall demand, and what demand patterns may be expected in the future.

To address this need, the **Demand Intelligence** project was developed as an interactive Power BI dashboard that brings together historical demand analysis, performance comparison, and demand forecasting.

The dashboard is organized around three connected perspectives:

Overview → Performance → Forecast

This structure allows users to first understand overall demand patterns, then identify the strongest store and product contributors, and finally examine expected future demand and forecast uncertainty.

Project Objectives

The analysis was designed to achieve the following objectives:

1. Understand Historical Demand Performance

Analyze demand trends over time to identify:

- Overall demand performance
- Year-over-year growth
- Monthly demand patterns
- Short-term demand movement using rolling averages
- Seasonal changes in demand

This provides a foundation for understanding how demand has evolved across the available historical period.

2. Identify Key Demand Contributors

Evaluate store and product performance to determine:

- The highest-performing store
- The highest-performing product
- The contribution of the top three stores
- The contribution of the top ten products
- The concentration of demand among leading contributors

This helps determine whether demand is broadly distributed or concentrated among a relatively small number of stores or products.

3. Analyze Demand Concentration

Compare leading contributors with the remaining business to understand the distribution of demand.

The analysis examines:

- Top store contribution versus remaining stores
- Top product contribution versus remaining products

This provides additional context beyond rankings by showing how dependent overall demand may be on top-performing contributors.

4. Forecast Future Demand

Extend the historical analysis by examining expected demand over the next forecast period.

The forecasting analysis focuses on:

- Expected future demand
- Monthly forecast patterns
- Forecast horizon
- Confidence interval and forecast uncertainty

The purpose is not to predict demand with absolute certainty, but to provide a structured view of expected future patterns based on the available historical data.

Key Business Questions

The project was designed to answer the following questions.

Overview

- How has total demand changed over time?
- What is the overall year-over-year growth trend?
- Which months typically experience higher or lower demand?
- How does short-term demand compare with the broader trend?

Performance

- Which store contributes the most to total demand?
- Which product generates the highest demand?
- How much demand is contributed by the top three stores?
- How much demand is contributed by the top ten products?
- Is demand concentrated among a small number of contributors?

Forecast

- What demand pattern is expected in the forecast period?
- Which months are expected to have higher or lower demand?
- How far into the future does the forecast extend?
- How much uncertainty should be considered when interpreting the forecast?

Expected Business Value

The completed dashboard provides a structured demand analysis that moves from understanding historical performance to identifying key contributors and evaluating expected future demand.

The analytical flow is:

Historical Demand → Performance Contributors → Demand Forecast

This structure allows stakeholders to use the report for multiple levels of analysis:

- Monitor overall demand performance
- Identify important stores and products
- Understand demand concentration
- Recognize recurring seasonal patterns
- Support forward-looking demand discussions

The dashboard is intended to support **demand monitoring and planning**, while the forecast should be interpreted as an estimate based on historical patterns rather than a guaranteed future outcome.

3. Data & Preparation

Data Source

The Demand Intelligence dashboard was built using a retail demand dataset structured in a SQL Server database named **RetailDemandDB**.

The analysis is based on four primary tables:

- **DimDate**
- **DimItem**
- **DimStore**
- **FactDemand**

A staging table, **dbo.stg_Demand**, was also used during the data preparation process.

The data model supports analysis across three key perspectives:

Time → Store → Product

Data Model Structure

The dataset follows a star-schema approach, with **FactDemand** serving as the central fact table and the remaining tables providing descriptive context.

FactDemand

The **FactDemand** table contains the demand records used throughout the analysis.

It connects demand activity to the relevant:

- Date
- Store
- Product

The table serves as the foundation for demand trends, store performance, product performance, and growth analysis.

DimDate

The **DimDate** table supports time-based analysis, including:

- Monthly and yearly trends
- Year-over-year growth
- Rolling averages
- Seasonal patterns

MonthName was also sorted by **MonthNumber** to ensure months were displayed chronologically across the report.

DimStore

The **DimStore** table provides store information used for:

- Store rankings
- Top store analysis
- Store contribution
- Demand concentration

DimItem

The **DimItem** table provides product information used for:

- Product rankings
- Top product analysis
- Product contribution

- Demand concentration

Data Preparation Process

The data preparation workflow followed a structured process:

Source Data → SQL Server → Data Model → Power BI Measures → Dashboard

The data was organized in SQL Server before being connected to Power BI for modeling, analysis, and visualization.

The preparation process focused on creating a structure that could support reliable analysis across dates, stores, and products.

Power BI Data Model

After connecting the prepared data to Power BI, the tables were organized into a relational model consisting of:

- **FactDemand** as the central fact table
- **DimDate** for time analysis
- **DimStore** for store analysis
- **DimItem** for product analysis

This structure allows measures and visuals to respond consistently to filters and slicers.

Supporting tables were also created for the demand concentration visuals:

- **Product Concentration**
- **Product Concentration Axis**
- **Store Concentration**
- **Store Concentration Bar**

A dedicated **_Measures** table was used to organize DAX calculations and keep the model structured and easier to maintain.

Key Analytical Measures

The dashboard uses DAX measures to calculate the metrics displayed across the three pages.

These measures support:

- Total and average demand
- Year-over-year growth
- Rolling averages
- Store and product rankings

- Top contributor analysis
- Demand concentration
- Latest demand
- Monthly forecast patterns

The detailed calculations and DAX logic are discussed in the next chapter.

Date Filtering and Interaction

A date slicer was added to allow users to control the analysis period.

The slicer dynamically updates relevant visuals and measures, including:

- Total demand
- Growth rates
- Store and product rankings
- Demand trends
- Latest demand values

Report interactions were tested to ensure visuals responded correctly to date selections and filter context.

Data Preparation Outcome

The final model provided the foundation for the dashboard's three analytical areas:

Overview — Historical demand trends and growth patterns.

Performance — Store and product contributors and demand concentration.

Forecast — Expected future demand patterns and forecast uncertainty.

The final structure combined SQL Server data preparation, a fact-and-dimension model, Power BI relationships, DAX measures, supporting tables, and interactive date filtering

Chapter Summary

The data preparation phase established a structured analytical foundation for the Demand Intelligence dashboard.

The final model enabled the project to analyze demand consistently across:

Time → Store → Product

With the data model in place, the next stage focused on transforming the prepared data into meaningful business metrics using **DAX measures and time intelligence logic**.

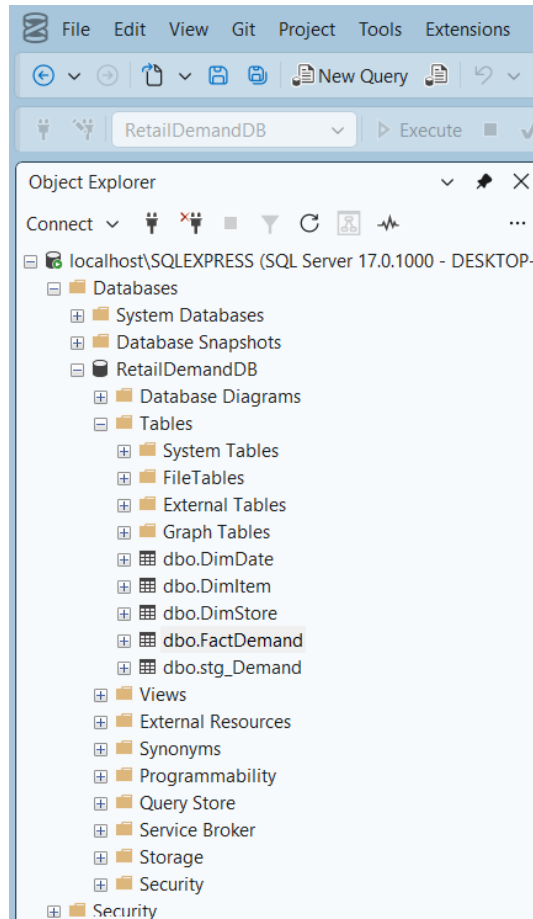


Figure 1— SQL Server

4. Data Modeling & DAX Measures

The Demand Intelligence dashboard uses a structured Power BI model and DAX measures to transform the prepared data into dynamic business metrics.

The measures respond to slicers, filters, and visual interactions, allowing the dashboard to update based on the selected analysis period.

The overall approach was:

Data Model → DAX Measures → Filter Context → Business Insights

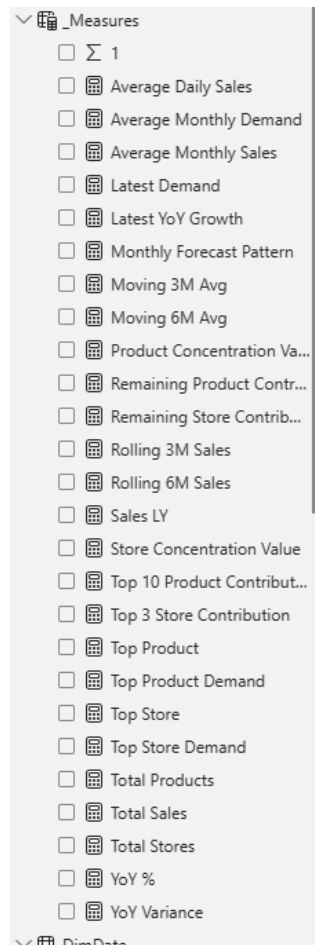


Figure 3— Measures

4.1 Core Demand Measures

The primary measure used throughout the dashboard is **Total Sales**, which serves as the project's main demand metric.

Total Sales =
SUM(FactDemand[Sales])

This measure forms the foundation for demand trends, store and product performance, growth calculations, and concentration analysis.

Additional measures were created to support the dashboard KPIs, including:

- Average Daily Sales
- Average Monthly Sales
- Total Stores
- Total Products

- Latest Demand

Although the source column is named **Sales**, it represents the primary demand metric used throughout the project.

4.2 Growth and Trend Analysis

Time intelligence measures were created to analyze how demand changed over time.

These include:

- **Sales LY**
- **YoY Variance**
- **YoY %**
- **Moving 3M Avg**
- **Moving 6M Avg**
- **Rolling 3M Sales**
- **Rolling 6M Sales**

The year-over-year measures compare current demand with the corresponding period from the previous year, while rolling averages help reduce short-term fluctuations and reveal broader demand trends.

These calculations support the historical demand analysis on the Overview page.

4.3 Store and Product Performance

Dynamic ranking measures were created to identify the strongest demand contributors.

Store Measures

- Top Store
- Top Store Demand
- Top 3 Store Contribution

Product Measures

- Top Product
- Top Product Demand
- Top 10 Product Contribution

The contribution measures calculate the share of total demand generated by the leading stores and products.

This allows the analysis to move beyond simple rankings and evaluate how dependent overall demand is on its top contributors.

4.4 Demand Concentration Analysis

Supporting DAX measures were created to compare leading contributors with the remaining stores and products.

These include:

- Store Concentration Value
- Remaining Store Contribution
- Product Concentration Value
- Remaining Product Contribution

The logic compares:

Top 3 Stores vs. Remaining Stores

and:

Top 10 Products vs. Remaining Products

This analysis helps determine whether demand is concentrated among a small group of contributors or broadly distributed across the business.

4.5 Latest Demand and Forecast Analysis

The **Latest Demand** measure was designed to return the most recent available demand value within the selected date range.

This ensures that the KPI responds dynamically when users change the date slicer.

The dashboard also includes a **Monthly Forecast Pattern** measure, which summarizes historical demand patterns by month and supports the seasonal analysis displayed on the Forecast page.

4.6 Measure Organization

All calculations were organized in a dedicated **_Measures** table to keep the Power BI model clean and easier to maintain.

The measures support five main analytical areas:

- Demand metrics
- Growth and trend analysis
- Store performance

- Product performance
- Forecast and seasonal analysis

5. Dashboard Design & Analytical Storytelling

The Demand Intelligence dashboard was designed to present more than individual metrics and charts. The goal was to create a structured analytical experience that guides users from understanding historical demand to identifying performance drivers and examining future demand expectations.

The dashboard follows a three-page storytelling framework:

Overview → Performance → Forecast

Each page answers a different business question while remaining connected to the overall demand story.

5.1 Design Philosophy

The dashboard was designed around four core principles:

Clarity

Each page focuses on a specific analytical objective. Visuals were selected based on the business question they needed to answer rather than simply filling available space.

The goal was to help users quickly understand:

- What the page is about
- Which metrics matter most
- What patterns are visible
- What insights can be drawn from the analysis

Consistency

A consistent visual identity was maintained across all three pages through:

- Clean white and light backgrounds
- Navy blue as the primary visual identity
- Purple accents for forecast-related analysis
- Consistent KPI cards and typography
- Rounded containers and structured spacing

- Customized icons aligned with each analytical section

This consistency allows the dashboard to feel like one connected analytical product rather than three separate reports.

Visual Hierarchy

Each page follows a structured information flow:

Page Title → KPI Summary → Main Analysis → Supporting Analysis → Key Insights

The layout allows users to review the dashboard at different levels, from a quick KPI scan to deeper analysis and business interpretation.

Purposeful Storytelling

Each page was designed to answer a specific question:

Page	Primary Question
Overview	What happened to demand?
Performance	What is driving demand?
Forecast	What might happen next?

5.2 Dashboard Page 1 — Demand Overview

Business Question

How has demand changed over time?

The Overview page establishes the historical foundation for the analysis before examining individual contributors or future demand.

KPI Summary

The top section provides a high-level view of:

- Total Demand
- Year-over-Year Growth
- Average Daily Demand
- Total Stores
- Total Products

These KPIs provide immediate context around demand scale, growth, and business coverage.

Demand Trend and Seasonality

The main demand trend combines monthly demand with a **3-month moving average**.

This helps users distinguish between short-term fluctuations and broader demand movement.

A separate monthly average visual highlights recurring seasonal patterns and makes it easier to identify months with stronger or weaker demand.

Yearly Performance

The yearly analysis combines annual demand with year-over-year growth.

This provides additional context by showing not only whether demand increased, but also **how quickly demand was changing**.

Demand Signals

A dedicated insights section summarizes the most important observations, including:

- Strongest growth period
- Growth slowdown
- Seasonal patterns
- Overall demand distribution

This helps bridge the gap between data visualization and business interpretation.



Figure 4— Demand Overview

5.3 Dashboard Page 2 — Performance

Business Question

Which stores and products are driving demand?

After establishing the overall demand trend, the Performance page examines the contributors behind the numbers.

Performance KPIs

The top section highlights:

- Top Store
- Top Product
- Top 3 Store Contribution
- Top 10 Product Contribution

These metrics immediately identify the strongest contributors while introducing the concentration analysis.

The page therefore answers both:

Who performs best?

and:

How dependent is overall demand on the leading contributors?

Store and Product Rankings

The ranking visuals compare demand across stores and products, allowing users to quickly identify:

- Top-performing contributors
- Relative performance
- Differences between leading and lower-performing contributors

Demand Concentration

The concentration analysis compares:

- **Top 3 Stores vs. Remaining Stores**
- **Top 10 Products vs. Remaining Products**

This moves the analysis beyond rankings by evaluating how broadly demand is distributed across the business.

The concentration visuals help answer:

Is demand heavily dependent on a small number of stores or products?



Figure 5— Performance

5.4 Dashboard Page 3 — Forecast

Business Question

What might happen next?

The Forecast page extends the analysis beyond historical performance and provides a forward-looking view of demand.

The page clearly distinguishes between:

- Historical demand
- Forecasted demand
- Forecast uncertainty

Forecast KPI Summary

The top section provides context through:

- Latest Demand
- 3-Month Average
- Forecast Horizon

- Confidence Level

These metrics help users understand recent demand activity before interpreting the projected outlook.

Demand Outlook

The central forecast visual combines:

- Historical demand
- Forecasted demand
- Confidence intervals
- Forecast start indicator

The forecast divider clearly separates actual results from projected values.

The confidence interval reinforces that future demand represents a range of possible outcomes rather than a guaranteed result.

Historical demand is observed. Future demand is estimated.

Monthly Forecast Pattern

The monthly forecast pattern provides an additional view of expected demand behavior across the year.

Based on historical seasonal patterns, it helps users identify:

- Stronger expected demand periods
- Lower demand periods
- Recurring seasonal behavior

Forecast Insights

The insight section summarizes the key forward-looking observations, including expected trends, seasonality, and forecast uncertainty.

This helps users interpret the forecast as a planning tool rather than simply reading a projected line.



Figure 6— Forecast

5.5 Cross-Page Storytelling

The three pages were intentionally designed as a connected analytical journey.

Page 1 — Overview

What happened?

Users begin by understanding historical demand, growth, and seasonal patterns.



Page 2 — Performance

What is driving demand?

Users then identify the stores and products contributing to overall demand.



Page 3 — Forecast

What might happen next?

Finally, users examine expected demand patterns and forecast uncertainty.

The complete analytical flow can be summarized as:

Understand the Past → Identify the Drivers → Prepare for the Future

This structure ensures that the forecast is supported by the historical patterns and performance analysis presented earlier in the dashboard.

6. Key Findings

The analysis identified several important patterns across historical demand, performance contributors, and future demand expectations.

These findings bring together the key insights from the Demand Overview, Performance, and Forecast analysis.

6.1 Demand Reached 47.70M Across the Historical Period

The business generated approximately **47.70M in total demand** between **2013 and 2017** across:

- **10 Stores**
- **50 Products**

This establishes the overall scale of demand analyzed in the project and provides the foundation for evaluating growth, performance, and future demand patterns.

6.2 Demand Continued to Grow, but Growth Momentum Moderated

Demand experienced its strongest year-over-year growth in **2014 at 15.04%**.

Growth remained positive in the following years but became more moderate and variable:

- **2015:** 4.39%
- **2016:** 8.60%
- **2017:** 3.64%

Although demand continued to increase, the lower growth rate in 2017 suggests that demand expansion had moderated compared with the stronger growth observed earlier in the historical period.

Demand remained on an upward trajectory, but growth momentum slowed over time.

6.3 Demand Shows Recurring Seasonal Patterns

The monthly analysis revealed recurring variation throughout the year.

Demand generally strengthened during the middle portion of the year, with **July recording the highest average monthly demand at approximately 1.0M.**

This indicates that demand is influenced by recurring seasonal behavior rather than remaining consistent throughout the year.

Seasonality is an important factor when monitoring demand and preparing for future periods.

6.4 Store 02 Is the Leading Store Contributor

The store performance analysis identified **Store 02** as the highest-performing location, generating approximately:

6.1M in Total Demand

Store 02 was the largest individual contributor, but the broader store network continued to generate most of the overall demand.

This highlights the importance of evaluating both individual performance leaders and the wider contribution of the store network.

6.5 Product 15 Is the Leading Product Contributor

The product performance analysis identified **Product 15** as the highest-performing product, generating approximately:

1.61M in Total Demand

While Product 15 leads the product portfolio, overall demand is supported by multiple products rather than being dependent on a single item.

6.6 Demand Is Diversified Across Stores

The store concentration analysis found that:

- **Top 3 Stores:** 36.5% of total demand
- **Remaining 7 Stores:** 63.5% of total demand

Although the leading stores are important contributors, the majority of demand comes from the broader store network.

The business is not heavily dependent on a small number of store locations.

This diversified store contribution provides greater stability than a demand base concentrated among only a few locations.

6.7 Demand Is Broadly Distributed Across Products

The product concentration analysis found that:

- **Top 10 Products:** 31.4% of total demand
- **Remaining 40 Products:** 68.6% of total demand

The majority of demand is generated by products outside the top-performing group.

The product portfolio has a broad demand base and is not heavily dependent on a small number of products.

6.8 Diversification Is a Key Strength

When store and product concentration are evaluated together, a consistent pattern emerges:

Dimension	Leading Group	Contribution	Remaining Contribution
Stores	Top 3 Stores	36.5%	63.5%
Products	Top 10 Products	31.4%	68.6%

Both analyses show that most demand comes from contributors outside the leading groups.

This suggests that overall demand is supported by a diversified base of stores and products, reducing dependency on individual contributors.

6.9 Historical Patterns Support Future Planning

The forecast analysis extends the historical demand pattern into a **12-month outlook**.

The projected pattern reflects the seasonal behavior observed in the historical data, providing useful context for future demand planning.

However, the confidence intervals widen further into the forecast horizon, indicating increasing uncertainty over time.

Forecasts should be used as planning guidance rather than guaranteed outcomes.

Overall Key Insight

The analysis shows a business with a strong and diversified demand base.

Historical demand grew across the available period, but growth momentum became more moderate in later years. Demand also follows recurring seasonal patterns that provide useful context for planning.

At the contributor level, both store and product demand are broadly distributed, reducing dependency on individual locations or products. The forecast provides a useful forward-looking view, but future demand should be interpreted alongside forecast uncertainty.

The overall analytical story can be summarized as:

Strong Historical Demand → Moderating Growth → Seasonal Patterns → Diversified Contributors → Forward-Looking Uncertainty

These findings provide the foundation for the business recommendations discussed in the next chapter.

7. Recommendations

Based on the key findings, the following recommendations can help support demand monitoring, planning, and future decision-making.

7.1 Plan Ahead for Seasonal Demand Patterns

The analysis shows recurring seasonal variation, with demand generally strengthening during the middle of the year.

Historical demand patterns should be considered when planning:

- Inventory availability
- Product replenishment
- Store capacity
- Operational resources

Recommendation

Use historical seasonal patterns to prepare for expected demand peaks rather than reacting only after demand increases.

7.2 Monitor Growth Momentum Alongside Total Demand

Total demand continued to grow throughout the historical period, but the pace of growth became more moderate in later years.

Monitoring total demand alone may hide important changes in performance.

Recommendation

Track both:

- **Total Demand**
- **Year-over-Year Growth**

This provides a clearer understanding of whether demand is:

- Accelerating
- Stabilizing
- Slowing

Strong demand levels do not always mean strong growth momentum.

7.3 Maintain Visibility Across the Entire Store Network

Store 02 is the highest-performing location, but the remaining stores generate **63.5% of total demand**.

Focusing only on the top-performing stores could overlook important changes across the broader network.

Recommendation

Monitor store performance across the entire network to identify:

- Emerging high-performing stores
- Declining store demand
- Changes in store rankings
- Shifts in demand concentration

7.4 Monitor the Broader Product Portfolio

Although Product 15 is the leading product, the remaining products generate **68.6% of total demand**.

The broader product portfolio therefore plays a significant role in overall demand performance.

Recommendation

Monitor product performance beyond the highest-ranking items by tracking:

- Product demand trends
- Changes in product rankings
- Emerging high-demand products
- Declining product demand

Demand planning should consider the full product portfolio rather than focusing only on top-performing products.

7.5 Track Demand Concentration Over Time

The current analysis shows a relatively diversified demand base across both stores and products.

However, concentration can change as demand patterns evolve.

Recommendation

Regularly monitor:

- Top Store Contribution
- Top Product Contribution
- Store Demand Concentration
- Product Demand Concentration

Increasing concentration could indicate greater dependency on a small number of contributors and potential exposure to performance changes.

7.6 Compare Actual Demand Against Forecasts

The forecast provides a useful outlook, but projected demand should be regularly compared with actual performance as new data becomes available.

Recommendation

Create a continuous review process:

Forecast → Actual Demand → Variance Analysis → Updated Planning

This would allow the business to identify significant forecast deviations and adjust planning assumptions when necessary.

7.7 Use Forecasts as Planning Guidance

The forecast confidence interval widens further into the future, indicating increasing uncertainty over longer forecast horizons.

Recommendation

Use forecasts differently depending on the planning horizon:

Short-Term Planning

- **Inventory planning**
- **Resource allocation**
- **Operational preparation**

Long-Term Planning

- **Scenario planning**
- **Capacity discussions**
- **Strategic considerations**

Forecasts should support decision-making, but they should not be treated as guaranteed outcomes.

8. Limitations & Next Steps

While the Demand Intelligence dashboard provides useful insights into historical demand, performance contributors, and future demand patterns, several limitations should be considered.

8.1 Dataset Scope

The analysis is based on historical data from **2013 to 2017**, covering **10 stores and 50 products**.

The dataset does not include additional factors that may influence demand, such as:

- Promotions and pricing
- Inventory availability and stockouts
- Customer behavior
- External market conditions

Next Step

Future versions could include additional business and operational data to provide a deeper understanding of the factors driving demand.

8.2 Forecasting Limitations

The forecast is based primarily on historical demand patterns and seasonality.

Future demand may be affected by events or conditions not represented in the historical data, and forecast uncertainty increases further into the forecast period.

Next Step

Future improvements could include:

- Actual vs Forecast analysis
- Forecast error metrics
- Regular forecast updates
- Testing alternative forecasting methods

The forecast should be used as planning guidance rather than a guaranteed prediction.

8.3 Limited Demand Driver Analysis

The dashboard identifies **when demand changes and where demand comes from**, but the available data provides limited information about **why demand changes**.

For example, changes in demand cannot be directly linked to promotions, pricing, inventory, or customer behavior.

Next Step

Additional data such as pricing history, promotions, inventory levels, and customer information could help move the analysis beyond descriptive insights and provide a better understanding of demand drivers.

8.4 Future Dashboard Opportunities

The current dashboard focuses on:

- Historical demand trends
- Store and product performance
- Future demand outlook

Future development could include:

- Store and product-level trend analysis
- Forecast accuracy monitoring
- Demand anomaly detection
- Automated alerts for significant demand changes

Overall Limitation and Future Direction

The dashboard provides a strong analytical foundation but is primarily limited by the scope of the available historical data.

The next stage of development should focus on:

Adding more demand drivers → Improving forecast validation → Expanding ongoing performance monitoring

These improvements could help evolve the project from a historical reporting dashboard into a more advanced demand planning and decision-support solution.

9. Conclusion

The **Demand Intelligence Dashboard** was developed to transform historical retail demand data into a clear and interactive analytical solution.

Using **SQL Server and Power BI**, the project analyzed demand across time, stores, and products while also providing a forward-looking forecast.

The analysis identified several important patterns, including:

- Strong overall historical demand growth with moderating momentum in later years
- Recurring seasonal demand patterns
- Store 02 and Product 15 as the leading individual contributors
- A diversified demand base across stores and products
- Increasing forecast uncertainty over longer planning horizons

The dashboard brings these insights together through three connected analytical areas: **Demand Overview, Performance, and Forecasting**.

Overall, the project demonstrates how data modeling, DAX calculations, and effective dashboard design can transform raw demand data into meaningful business insights.

The final solution provides a foundation for monitoring demand performance, identifying key contributors, recognizing seasonal patterns, and supporting more informed planning decisions.

While the analysis is limited by the available dataset and forecast assumptions, future improvements could incorporate additional demand drivers and forecast validation to strengthen the solution further.

Ultimately, this project demonstrates the value of combining **data preparation, analytical modeling, and business-focused storytelling** to create a practical demand intelligence solution.